THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 2: **Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If α, β are zeroes of $x^2 - 4x + 3$, then $\alpha^3 + \beta^3$ is: 1

(a) 28 **(b)** 100 **(c)** 64 **(d)** 40

2. If α, β are zeroes of $x^2 - 5x + 6$, then $\alpha^2\beta + \alpha\beta^2$ is: 1

(a) 30 **(b)** 11 **(c)** 36 **(d)** 25

3. If α, β are zeroes of $x^2 - 11x + 18$, then $|\alpha - \beta|$ is: 1

(a) 7 **(b)** 8 **(c)** 11 **(d)** 14

4. If α, β are zeroes of $x^2 - 6x + 8$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is: 1

(a) $\frac{5}{2}$ **(b)** $\frac{9}{2}$ **(c)** $\frac{3}{4}$ **(d)** 20

5. For the cubic with zeroes $1, 2, 3$, $\alpha\beta + \beta\gamma + \gamma\alpha$ is: 1

(a) 11 **(b)** 6 **(c)** 36 **(d)** -11

6. If the zeroes of $x^2 - 10x + k$ differ by 4, then k is: 1

(a) 42 **(b)** 29 **(c)** 21 **(d)** 84

7. If $x = 2$ is a zero of $x^2 - 9x + 14$, the other zero is: 1

(a) 12 **(b)** -7 **(c)** 11 **(d)** 7

8. If α, β are zeroes of $x^2 - 6x + 8$, then $\alpha^3 + \beta^3$ is: 1

(a) 360 **(b)** 216 **(c)** 120 **(d)** 72

9. **Assertion (A):** A quadratic polynomial whose graph is a parabola cutting the x-axis at two distinct points has two real zeros. 1

Reason (R): The zeros of a polynomial $p(x)$ are the x-coordinates of the points where the graph of $y = p(x)$ meets the x-axis.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The sum of the zeros of $2x^2 - 8x + 6$ is 4. 1

Reason (R): For $ax^2 + bx + c$ with zeros α, β, the sum of the zeros is $\frac{c}{a}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If α, β are zeroes of $x^2 - 4x + 3$, form a quadratic polynomial whose zeroes are α^2 and β^2. 2

12. If the sum of the zeroes of $x^2 - 6x + k$ is equal to their product, find k. 2

13. If one zero of $x^2 - 8x + k$ is 3 times the other, find k. 2

14. If α, β are zeroes of $x^2 - 5x + 6$, form a quadratic polynomial whose zeroes are α^2 and β^2. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find a quadratic polynomial whose zeroes are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, where α, β are zeroes of $x^2 - 3x + 2$. 3

16. The zeroes of $x^2 - 10x + c$ are α and β with $\alpha - \beta = 4$. Find c. 3

17. If α, β, γ are the zeroes of $x^3 - 6x^2 + 11x - 6$, find $\alpha^2 + \beta^2 + \gamma^2$. 3

OR

Find a quadratic polynomial whose zeroes are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, where α, β are zeroes of $x^2 - 5x + 6$. 3

18. The zeroes of $x^2 - 12x + c$ are α and β with $\alpha - \beta = 6$. Find c. 3

OR

If α, β, γ are the zeroes of $x^3 - 7x^2 + 14x - 8$, find $\alpha^2 + \beta^2 + \gamma^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Meera is designing a parabola-shaped flower bed in her school garden. She models its boundary by the graph of $y = p(x) = x^2 - 8x + 15$, where x is measured in metres along the ground. 5

(i) Find the zeros of $p(x)$.

(ii) Find the sum and the product of the zeros.

(iii) How many times does the graph of $y = p(x)$ meet the x-axis, and at which points?

(iv) Form a quadratic polynomial whose zeros are the reciprocals of the zeros of $p(x)$.

20. The zeroes of $x^3 - 9x^2 + 23x - 15$ are in arithmetic progression. Find them. 5

OR

If α, β are the zeroes of $x^2 - 2x - 3$, form a quadratic polynomial whose zeroes are $\alpha + \beta$ and $\alpha\beta$. 5

END OF QUESTION PAPER — Total: 40 Marks